

Soroush Mahdi

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Education

Master of Science in Artificial Intelligence

Amirkabir University of Technology (Tehran Polytechnic)

Ranked **7th** among 50 M.Sc. students

Cumulative **GPA: 4/4**

Tehran, Iran

Sep 2020 – Sep 2023

Bachelor of Computer Engineering

Bu-Ali Sina University

Cumulative GPA: 3.17

Hamedan, Iran

Sep 2016 – Sep 2020

Research Interests

Vision-Language Models (VLMs)

Computational Neuroscience

Trustworthy ML

Computer Vision

Deep Learning Theory

Generative AI

Publications

1. **Mahdi, S.**, Ayar, F., Javanmardi, E., Tsukada, M., Javanmardi, M., "[Evict3R: Training-Free Token Eviction for Memory-Bounded Streaming Visual Geometry Transformers](#)". Submitted to ICRA 2025.
2. **Mahdi, S.**, Nabati, S., Dehghanian, Z., Amirmazlaghani, M., "[MemLoss: Enhancing Adversarial Training with Momentum-Based Memory](#)". Submitted to Applied Soft Computing.

Honors & Awards

- **1st** Place in [The Iran Annual AI Award](#), **focusing on Brain abnormality diagnosis using MRI DICOM images**, Cognitive Sciences and Technologies Council, Tehran, Iran, 2024.
- **2nd** Place in [The National fNIRS Data Analysis Competition](#) focusing on brain-computer interface systems to help patients with movement disorders, National Brain Mapping Laboratory, Tehran, Iran, 2022.
- Ranked **61st Among More Than 15,000 Participants** in Iranian University Entrance Exam For Masters In Artificial Intelligence, Iran, 2020.
- Ranked **36th Among More Than 15,000 Participants** in Iranian University Entrance Exam For Masters In Algorithms and Computation, Iran, 2020.
- Ranked **22nd** Place in [National Collegiate Scientific Olympiad in Computer Engineering](#), Iran National Organization of Educational Testing, Tehran, Iran, 2020.

Academic Experiences

Research Assistant

Autonomous & Intelligent Systems Lab (AISL), Amirkabir University of Technology

Under the supervision of Mahdi Javanmardi

June 2025 – Present

Tehran, Iran

- Working on 3D Computer Vision and Vision Language Models.
- Developed [Evict3r](#), a novel token eviction method for deep foundation SFM models, improving efficiency in large-scale scene reconstruction and reducing computational overhead.

Thesis Based M.Sc. Student and Research Assistant

Statistical Data Analysis Laboratory

Under the supervision of Maryam Amirmazlaghani

June 2021 – Present

Tehran, Iran

- Thesis title: An approach for improving the robustness of deep neural network image classifiers against adversarial examples.

Work Experiences

Data Scientist

Iran Credit Scoring

July 2024 – June 2025

Tehran, Iran

- Working as a data scientist in the R&D team.

Computer Vision Researcher

HARA.ai

Jan 2022 – Jan 2023

Tehran, Iran

- Specialized in face anti-spoofing and liveness detection using deep learning techniques.
- Developed blink detection and blink counter models using semi-supervised approaches.

Academic Projects

- **Robust Supervised contrastive learning** - [Github link](#)

Implemented adversarial training coupled with supervised contrastive learning using PyTorch.

- **PSO-SGD Hybrid Optimizer** - [Github link](#)

Developed a custom PyTorch optimizer by designing an optimization algorithm that combines SGD and PSO.

- **multivariate HMM** - [Github link](#)

Implemented a range of algorithms for Hidden Markov Models (HMMs) in the multivariate case from scratch.

Teaching Assistantships

Statistical Machine Learning - AmirKabir University of Technology

Instructor: Maryam Amir mazlaghani (Assoc. Prof.) - Graduate Course

Tehran, Iran

Spring 2023

Machine Vision - AmirKabir University of Technology

Instructor: Reza Safabakhsh (Prof.) - Graduate Course

Tehran, Iran

Fall 2022

Other Courses: Algorithm Design(2020), Artificial intelligence & Expert Systems(2019), Data Structures(2018)

Professional Skills

- **Programming Languages:**

Python, C++, R

- **Libraries/Frameworks:**

PyTorch, NumPy, Pandas, OpenCV, scikit-learn, Tensorflow, JAX

- **Tools:**

Git, LATEX, Visual Studio Code

- **Languages:**

English: TOEFL iBT Score: 103/120 (Reading: 30/30, Listening: 29/30, Speaking: 21/30, Writing: 23/30)

Persian: Native

Major Courses

Machine Vision	19.9/20	Stochastic Processes	19.28/20
Machine Learning	18.6/20	Neural Networks	16.8/20
Optimization	16.21/20	Computational Intelligence	20/20

References

Mahdi Javanmardi (Asst. prof.)	mjavan@aut.ac.ir	Homepage
Maryam Amir Mazlaghani (Assoc. Prof.)	mazlaghani@aut.ac.ir	Homepage
Mir Hossein Dezfoulan (Asst. Prof.)	dezfoulan@basu.ac.ir	Homepage
More references are available upon request.		